The Dirac delta function

We already understand that a step function is a function that changes value suddenly at a particular point. The value of the unit step function, or Heaviside function, changes value from 0 to 1 exactly at the point $t = c$.

$$u_c(t) = \begin{cases} 0 & 0 \leq t < c \\ 1 & t \geq c \end{cases}$$

The **Dirac delta function** (which is actually less like a function and more like a distribution) also models a sudden change. It's particularly useful for modeling the application of a very large force over a very short time.

Think about the force of a golf club striking a golf ball, or a hammer striking a nail as examples of extreme force that's applied almost instantaneously. These are the kinds of functions that are best handled using the Dirac delta function.

Properties of the Dirac delta function

There are three important properties of the Dirac delta function δ ("delta") that we need to understand. First, the function has a zero value everywhere except at $t = c$.

$$\delta(t - c) = 0 \text{ for } t \neq c$$

Think again about a golf club striking a golf ball. No force at all is being applied to the golf ball, except at that fraction of a second when $t = c$. So the function will have a zero value everywhere outside of that instant. But

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at that instant $t = c$, the force applied is so extreme that we almost think about it as being infinite.

Second, the integral of the Dirac delta function is 1, as long as the interval of integration includes the point $t = c$. If we think about Dirac delta as a distribution, then we can imagine all of the area under the distribution being condensed into the instant $t = c$. At that instant, we see all the area under the distribution, a total of 1. But outside of that instant, we see no force at all.

$$\int_{c-\epsilon}^{c+\epsilon} \delta(t-c)\, dt = 1 \text{ for } \epsilon > 0$$

Because the value of this integral is 1, we sometimes call $\delta(t-c)$ a "unit impulse." Third, multiplying the Dirac delta function by any function $f(t)$ gives us a non-constant model of this strong impulse function. And if we integrate that model, as long as our interval of integration includes $t = c$, then the value of the integral will be $f(c)$.

$$\int_{c-\epsilon}^{c+\epsilon} f(t)\delta(t-c)\, dt = f(c) \text{ for } \epsilon > 0$$

Applying the Laplace transform

The Laplace transform of the Dirac delta function is

$$\mathscr{L}(\delta(t-c)) = \int_0^\infty e^{-st}\delta(t-c)\, dt = e^{-cs} \text{ for } c > 0$$

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This transform is really what we've been building toward, so let's do an example where we find the Laplace transform, then solve an initial value problem with the Dirac delta function.

Example

Solve the initial value problem, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' + y = 2\delta(t - \pi)$$

From our table of Laplace transforms, we know

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(\delta(t - c)) = e^{-cs}$$

so the Laplace transform of the second order nonhomogeneous equation with a Dirac delta forcing function is

$$\mathscr{L}(y'') + \mathscr{L}(y) = \mathscr{L}(2\delta(t - \pi))$$

$$s^2Y(s) - sy(0) - y'(0) + Y(s) = 2e^{-\pi s}$$

Substitute initial conditions, then solve for $Y(s)$.

$$s^2Y(s) - s(1) - 0 + Y(s) = 2e^{-\pi s}$$

$$s^2Y(s) - s + Y(s) = 2e^{-\pi s}$$

$$s^2Y(s) + Y(s) = s + 2e^{-\pi s}$$

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$$(s^2+1)Y(s) = s + 2e^{-\pi s}$$

$$Y(s) = \frac{s}{s^2+1} + 2\left(\frac{1}{s^2+1}\right)e^{-\pi s}$$

To find the general solution $y(t)$, we'll take the inverse transform.

$$y(t) = \cos t + 2\sin(t-\pi)u(t-\pi)$$

Of course, we could also write this solution as

$$y(t) = \begin{cases} \cos t & 0 \le t < \pi \\ \cos t + 2\sin(t-\pi) & t \ge \pi \end{cases}$$

Relating the Dirac delta and unit step functions

The last point we want to make is that the Dirac delta function is actually the derivative of the unit step function.

We can prove this to ourselves if we start with what we already know about the Dirac delta function. We know that, if the interval of integration includes the point $t = c$, then the integral of the Dirac delta function will be 1. But if the interval doesn't include that one special point, then the integral will be 0. We can write this as

$$\int_{-\infty}^{t} \delta(u-c)\, du = \begin{cases} 0 & t < c \\ 1 & t > c \end{cases}$$

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But now we notice that this right side is the definition of the unit step function, or Heaviside function, so

$$\int_{-\infty}^{t} \delta(u-c)\ du = u_c(t)$$

From here, applying the Fundamental Theorem of Calculus to this equation shows that the Dirac delta function is the derivative of the unit step (Heaviside) function.

$$u_c'(t) = \frac{d}{dt}\left(\int_{-\infty}^{t} \delta(u-c)\ du\right) = \delta(t-c)$$

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